

VAM-T: Visual-Audio Temporal Mismatch Dataset for Video Temporal Grounding

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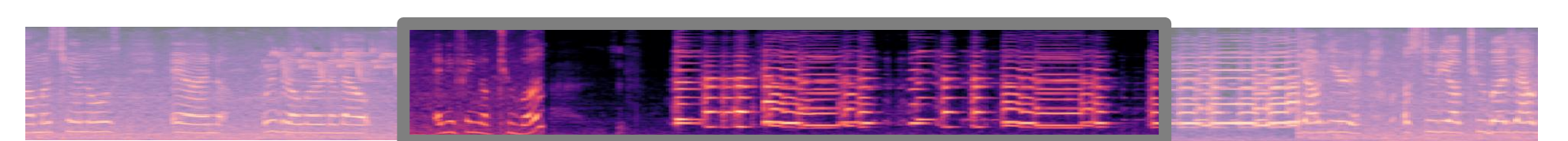
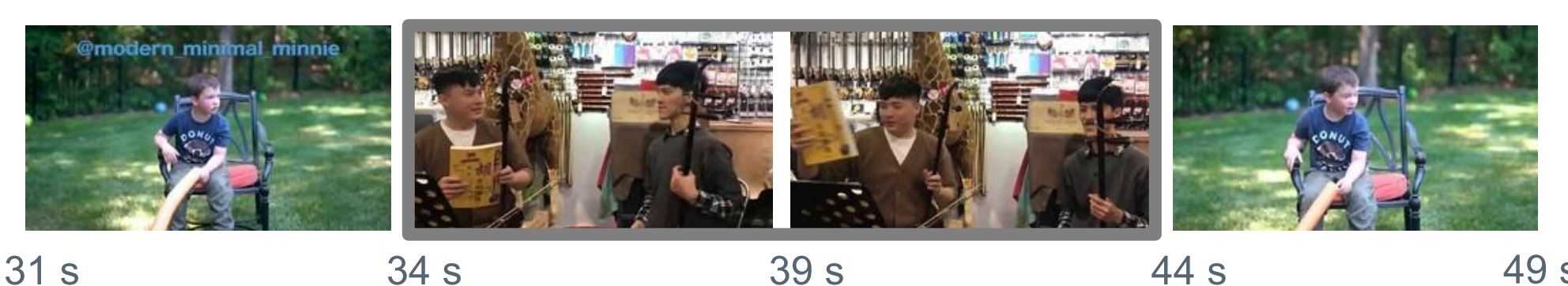
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01 Motivation

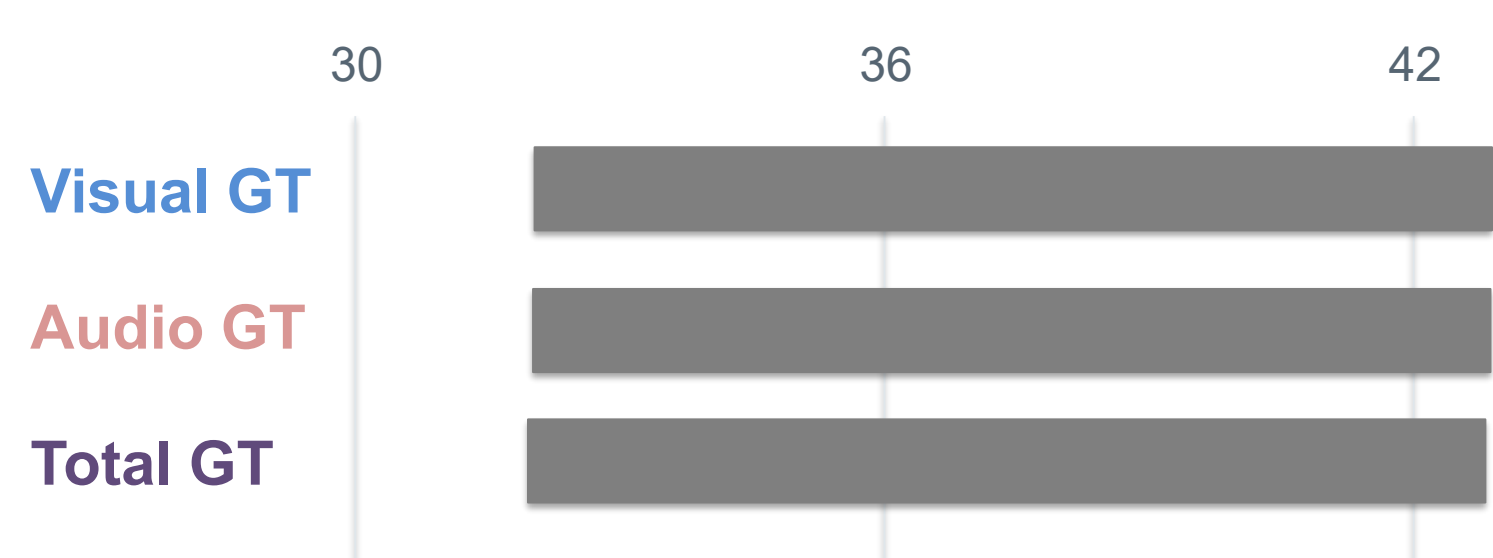
1) Video Temporal Grounding(VTG)

VTG is the task of identifying the start and end timestamps of the video segment most relevant to a given natural-language query.

Query: “When is the erhu being played?”



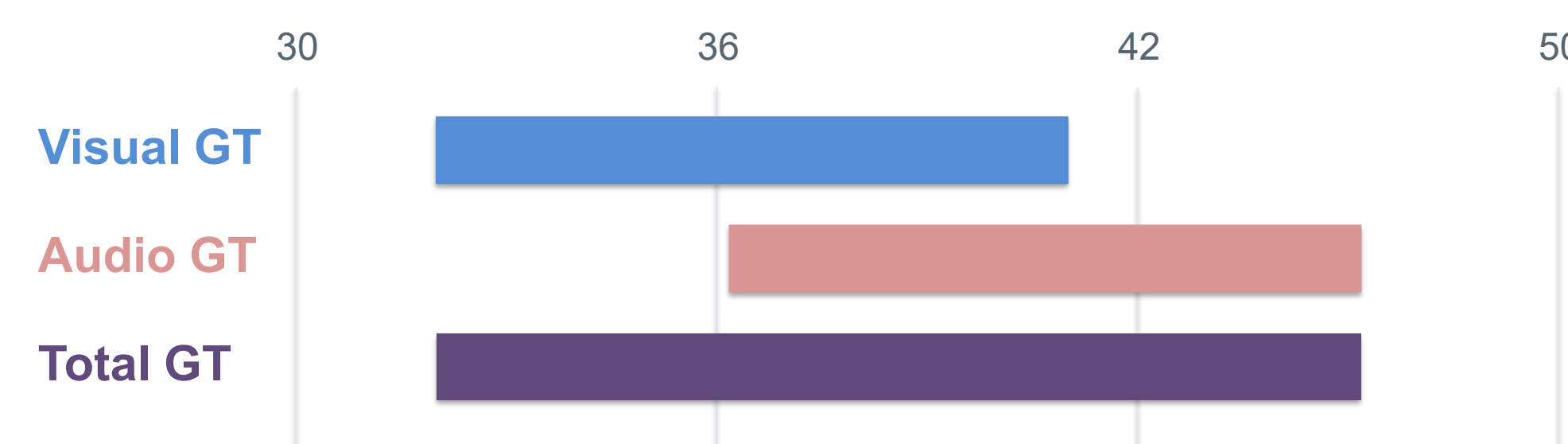
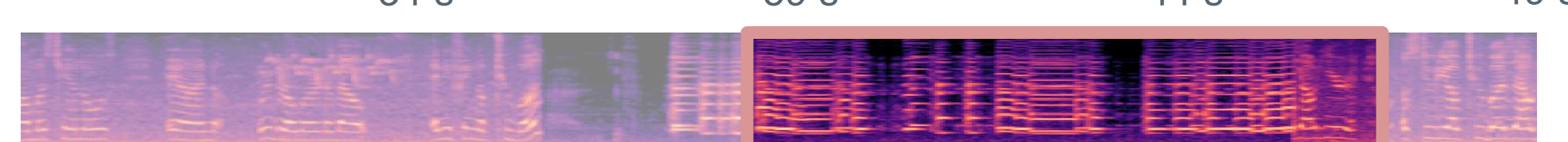
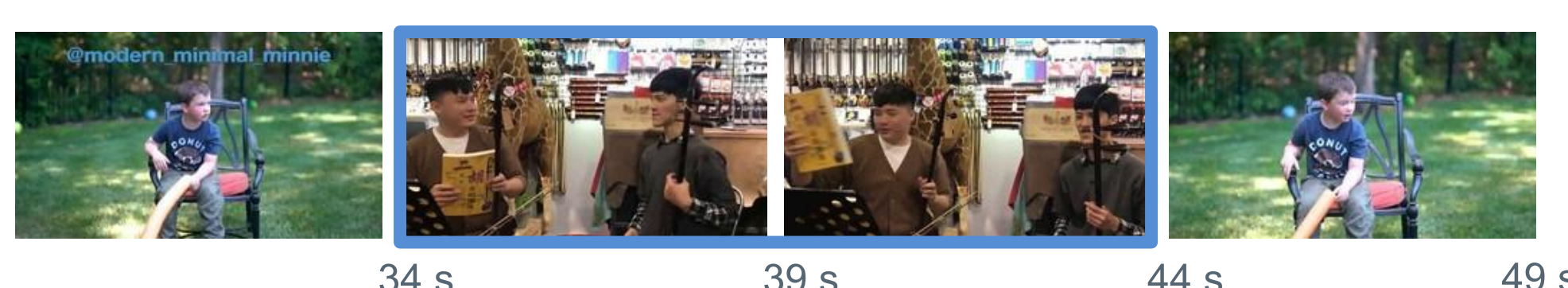
Answer: from 34s to 44s



2) Controlled Mismatch Reveals Missing Evidence

Aligned inputs can hide reliance on one modality.

Controlled temporal mismatch tests whether the model covers the complete audible and visible event.



Audio delay +4.18 s

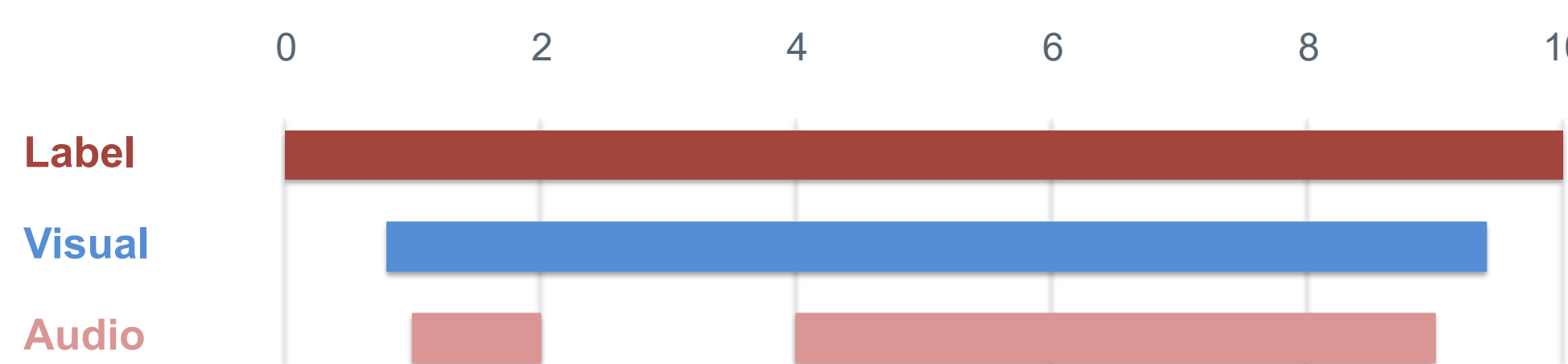
A visual-only prediction misses the delayed sound.

02 Data Quality & Synthesis

CLIP-LEVEL LABELS ARE NOT DENSE TEMPORAL TARGETS



An event may occupy only part of a 10-s VGGSound clip [1].



Illustrative timing: full-clip labels introduce temporal boundary error.

50,000 candidates → 23,214 validated clips

Human-aligned prompt/model + automated filtering · 46.4% retained

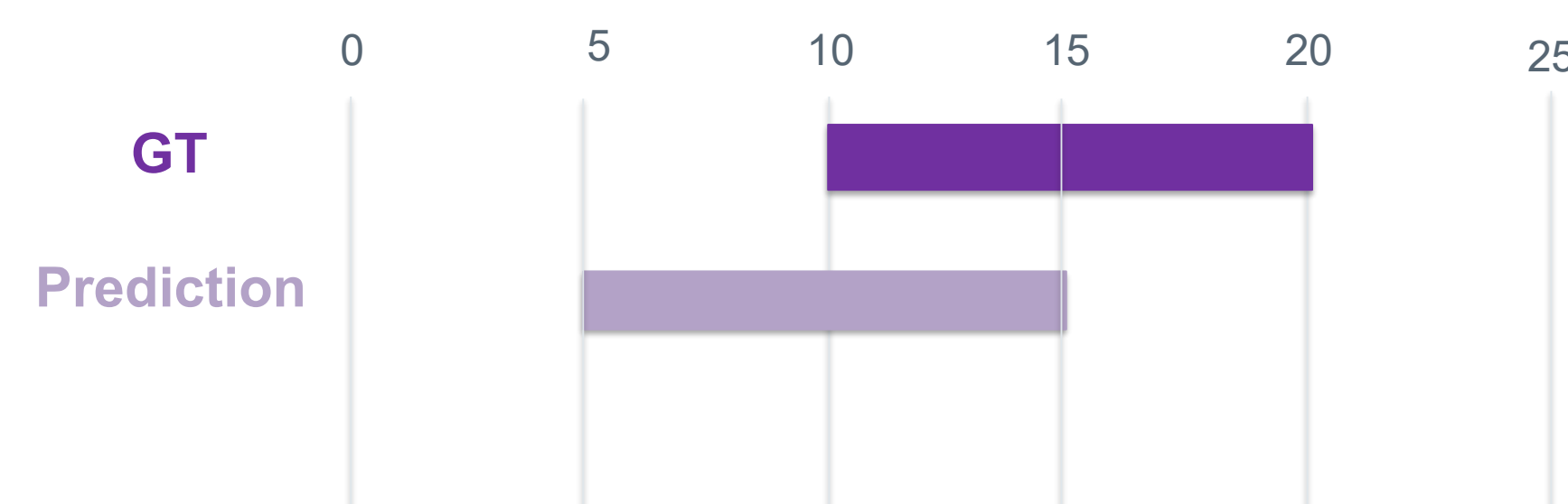
3,316 videos · 23,214 queries

Split	Videos	Queries
Train	2,686	18,805
Validation	315	2,207
Test	315	2,202

Targets: 5–10 s · nonzero shifts: 0.501–7.932 s
100-clip audit, 4 raters: Fleiss' κ = 0.686

03 Evaluation Metrics

1) Union Prediction · Modality-Specific Diagnosis



$$IoU = \frac{GT \cap Pred}{GT \cup Pred} = \frac{5}{15}$$

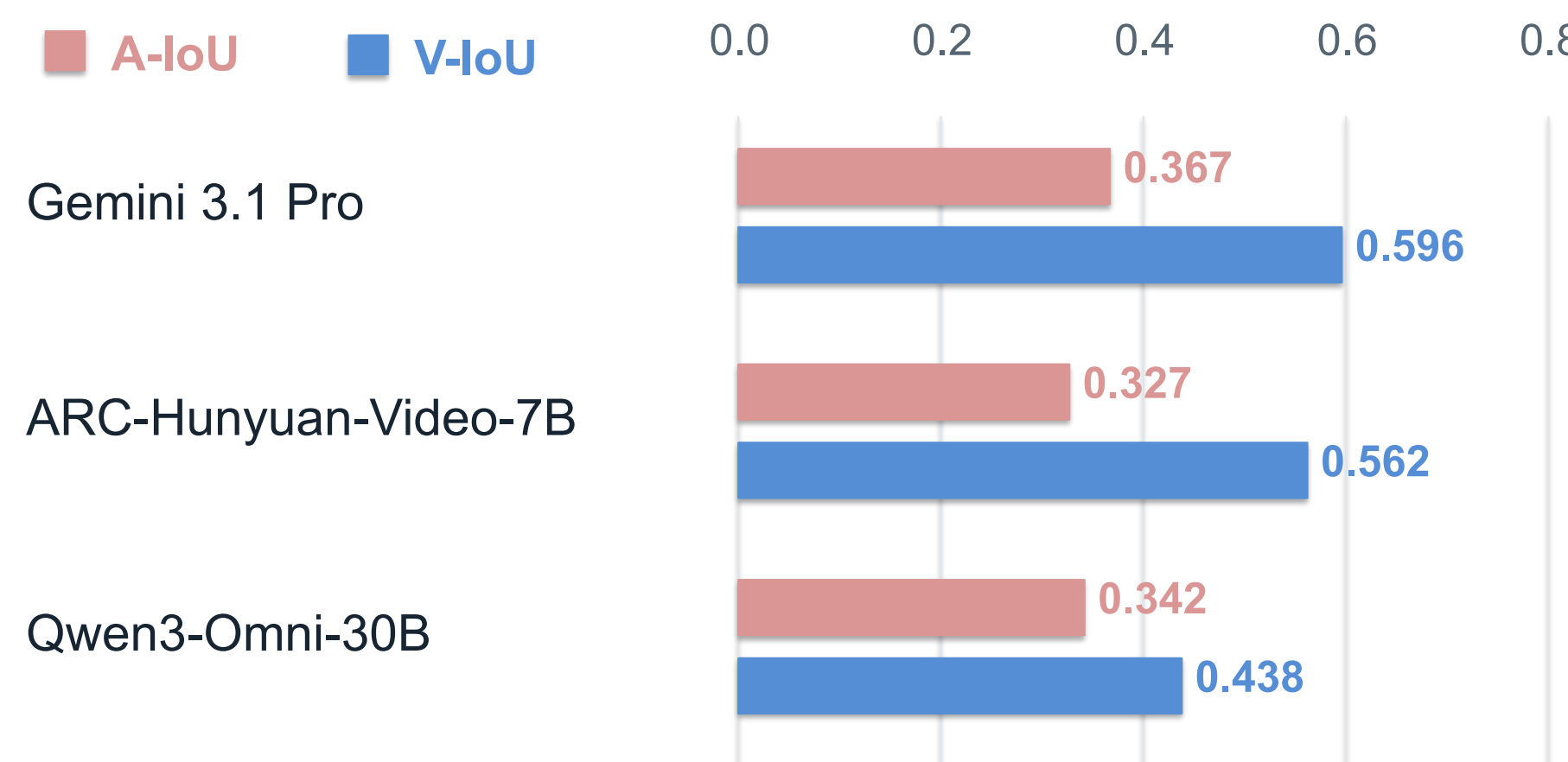
$V - IoU$ $A - IoU$

04 Mismatch Diagnosis

2) Audio-Delay Condition · A-IoU vs. V-IoU

Visual evidence remains easier to localize

Delayed audio-exclusive evidence is the main residual weakness.



Gemini 3.1 Pro · condition-wise mIoU

Aligned 0.633 → Audio delay 0.452

A high aligned score does not establish independent recovery of audio and visual evidence.

05 VAM-T-Qwen

QWEN2.5-OMNI-7B [2] · SFT + MODALITY-AWARE GRPO



1 Supervised Fine-Tuning

18,805 training queries · union supervision



2 Modality-Aware GRPO

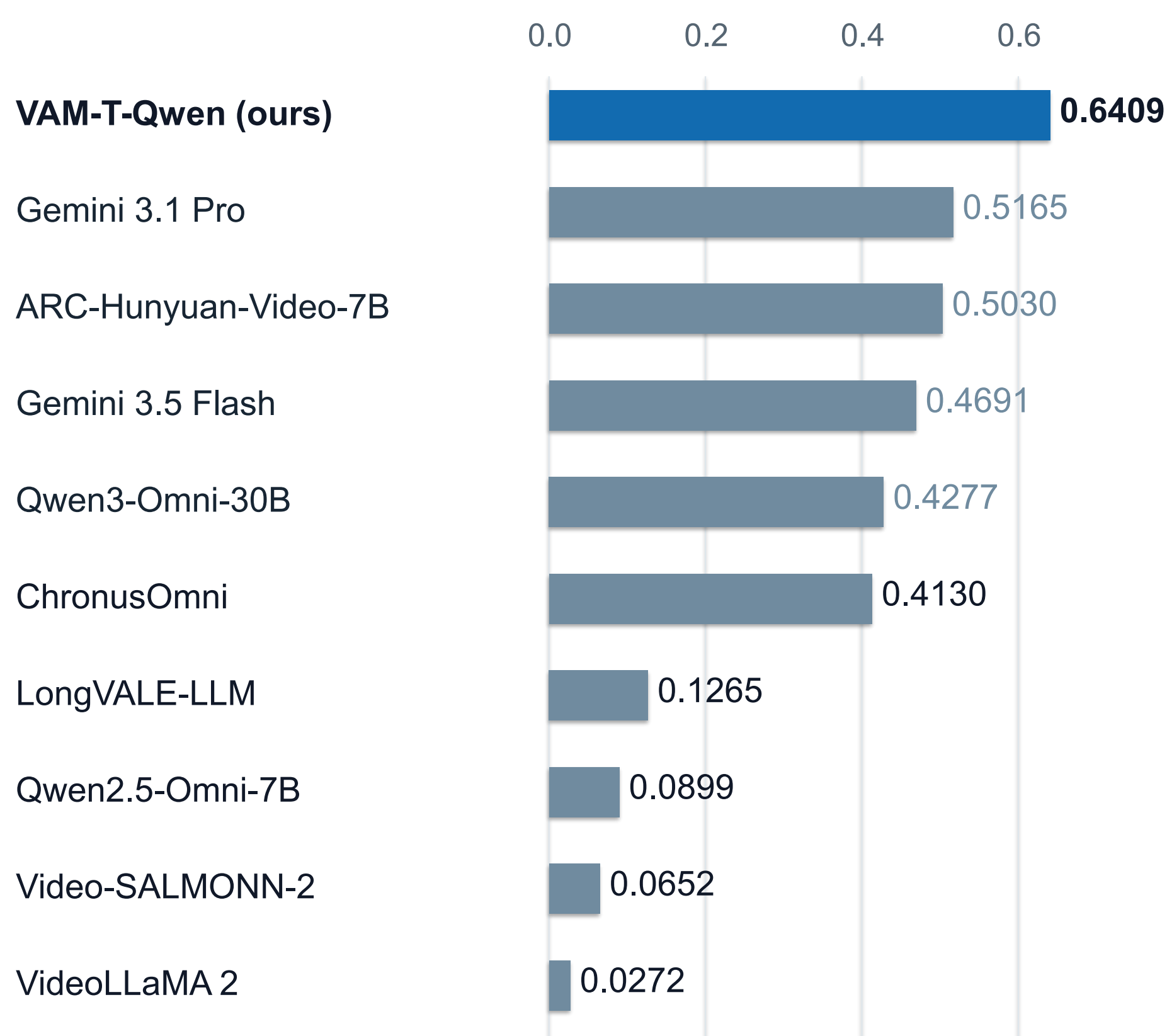
12,536 shifted queries · balanced lead/delay

$$r = 0.5 IoU_T + 0.5 \min(IoU_A, IoU_V) - 0.3 |IoU_A - IoU_V|$$

Our modality-aware reward for GRPO [3]

06 Grounding Performance & Generalization

FINAL VAM-T-QWEN · TEST SET mIoU ↑



VAM-T-Qwen achieves the highest mIoU under the common evaluation protocol.

VAM-T-Qwen vs. strongest baseline

mIoU +0.1244 R@0.5 0.833

Gemini 3.1 Pro → VAM-T-Qwen Final SFT+GRPO model

R@0.3/0.5/0.7 = 0.911/0.833/0.445 · A-IoU/V-IoU = 0.596/0.663

ZERO-SHOT TRANSFER · DURATION-CONTROLLED TESTS

VAM-T-Qwen vs. Qwen2.5-Omni-7B base

Benchmark / metric	Score	Δ (pp)
LongVALE · mIoU	18.41	+5.31
ChronusAV · V2T mIoU	29.56	+19.78
ChronusAV · A2T mIoU	14.92	+5.62
DAVE · Accuracy	30.46	+1.03
ACVUBench · Accuracy	64.86	+1.59

07 Conclusions

VAM-T · TEMPORAL-MISMATCH GROUNDING

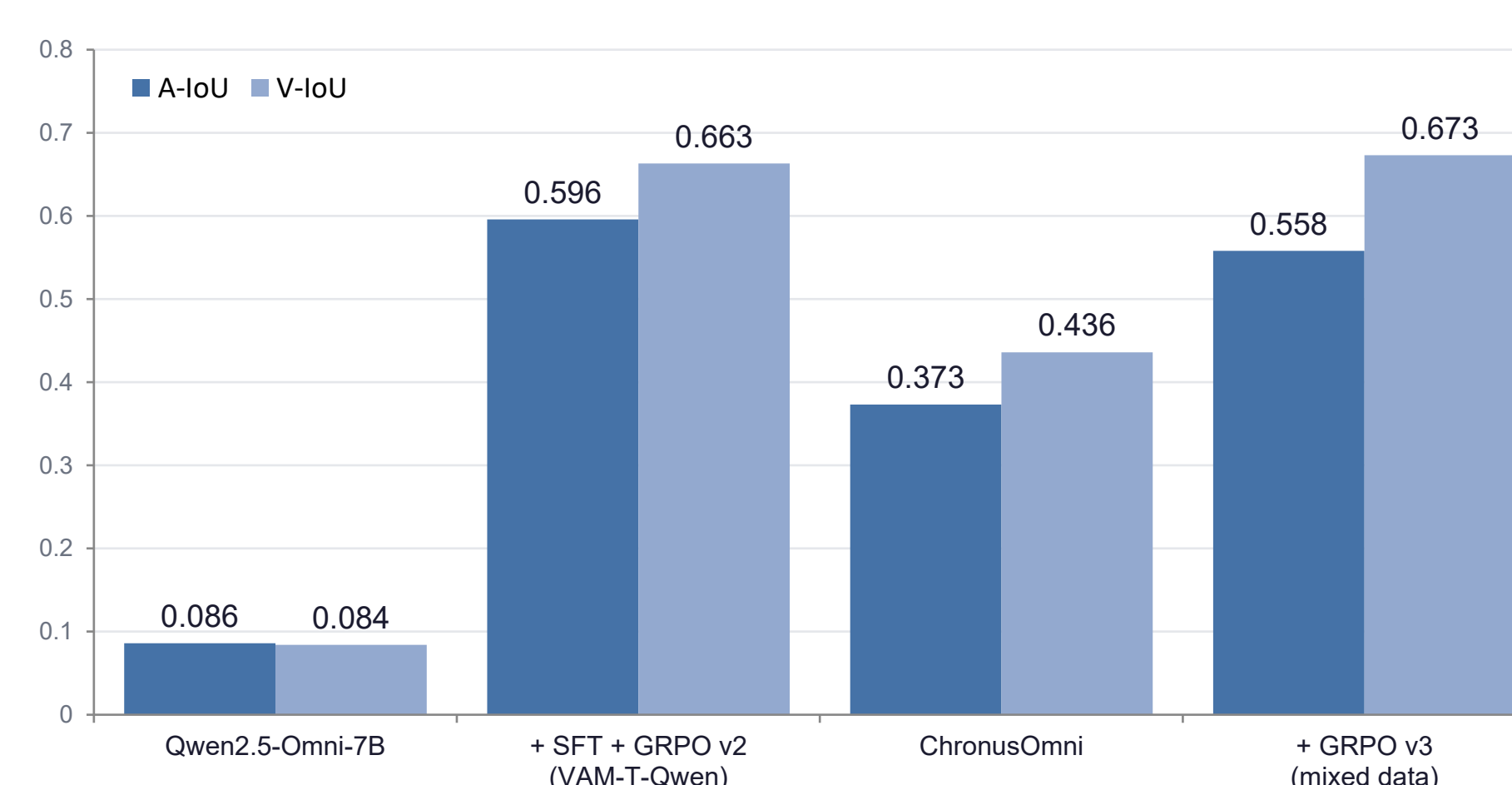
Independent A/V targets expose failures hidden by union timestamps.

MODEL SFT + modality-aware GRPO on Qwen2.5-Omni-7B.

RESULT 0.6409 mIoU; +0.1244 over the strongest baseline.

TRANSFER VAM-T-Qwen improves all five metrics across four external benchmarks in zero-shot evaluation.

Training improves localization for both modalities



LIMITATION The cross-backbone transfer effect attributable to VAM-T training has not yet been verified.

ONGOING EXPERIMENTS Cross-backbone transfer experiments with multiple training recipes and dataset configurations

Acknowledgment

This work was financially supported by the DGIST UGRP; computing resources were provided by NVIDIA and KISTI.

References

[1] H. Chen et al., “VGGSound,” ICASSP 2020. [2] J. Xu et al., “Qwen2.5-Omni,” arXiv:2503.20215, 2025. [3] Z. Shao et al., “DeepSeekMath,” arXiv:2402.03300, 2024.